

Design and Evaluation of a Drift-Aware Continuous Learning MLOps Framework for Financial Transaction Fraud Detection

Presented by Mashud Hasan (22299053), Nirnoy Biswas (24241185), Md. Sabbir Hossain (22299038),

Al Imran (231011035), Imtiaz Zaman Sami (23101551)

Supervisor: Md. Sabbir Ahmed | Co-Supervisor: Naima Tahsin Nodi

BRAC University



Abstract

Fraud models are usually trained once, but in production fraud patterns change (**concept drift**) and true labels arrive weeks late (**label delay**). We build and evaluate a **drift-aware continuous learning MLOps framework** that monitors a deployed model on delayed labels, retrains it on a schedule with an automatically chosen training window, and deploys a new model only if it passes a **promotion gate**. It is tested on three public datasets with label delays of 0 to 60 days.

Problem Statement

Existing studies test fraud models on **random splits**, assume **immediate labels** and trigger retraining with **drift detectors** on artificial drift. When should a real fraud model be retrained, and how can a bad model be kept out of production?

Research Objectives

1. Measure how a deployed fraud model decays and whether drift measures detect it.
2. Measure how label delay limits the benefit of retraining.
3. Compare scheduled and drift-triggered retraining fairly.
4. Build an MLOps framework with monitoring, retraining, window selection and a promotion gate, and test it with injected faults.
5. Replicate on three public datasets.

Literature Review

- **Dal Pozzolo et al.** [1]: fraud labels arrive late (verification latency), which limits how a model can be updated.
- **Menezes and Filho** [2]: fraud models on IEEE-CIS lose up to 40% F1 under natural drift; no retraining tested.
- **Shakil et al.** [3]: drift-triggered retraining for MLOps, tested only on injected drift with immediate labels.

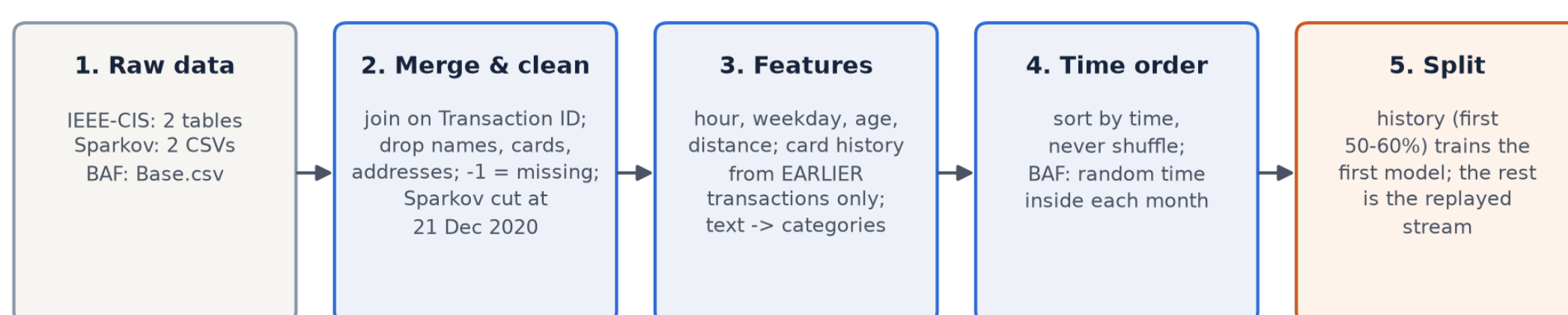
Gap: no fair test of retraining policies for fraud under real drift and label delay.

Data set

	IEEE-CIS	Sparkov	BAF
Type	Real e-commerce	Simulated cards	Bank accounts
Records	590,540	1,801,969	1,000,000
Time span	182 days	720 days	8 months
Fraud rate	3.50%	0.53%	1.10%
Features	431	16	29

Preprocessing

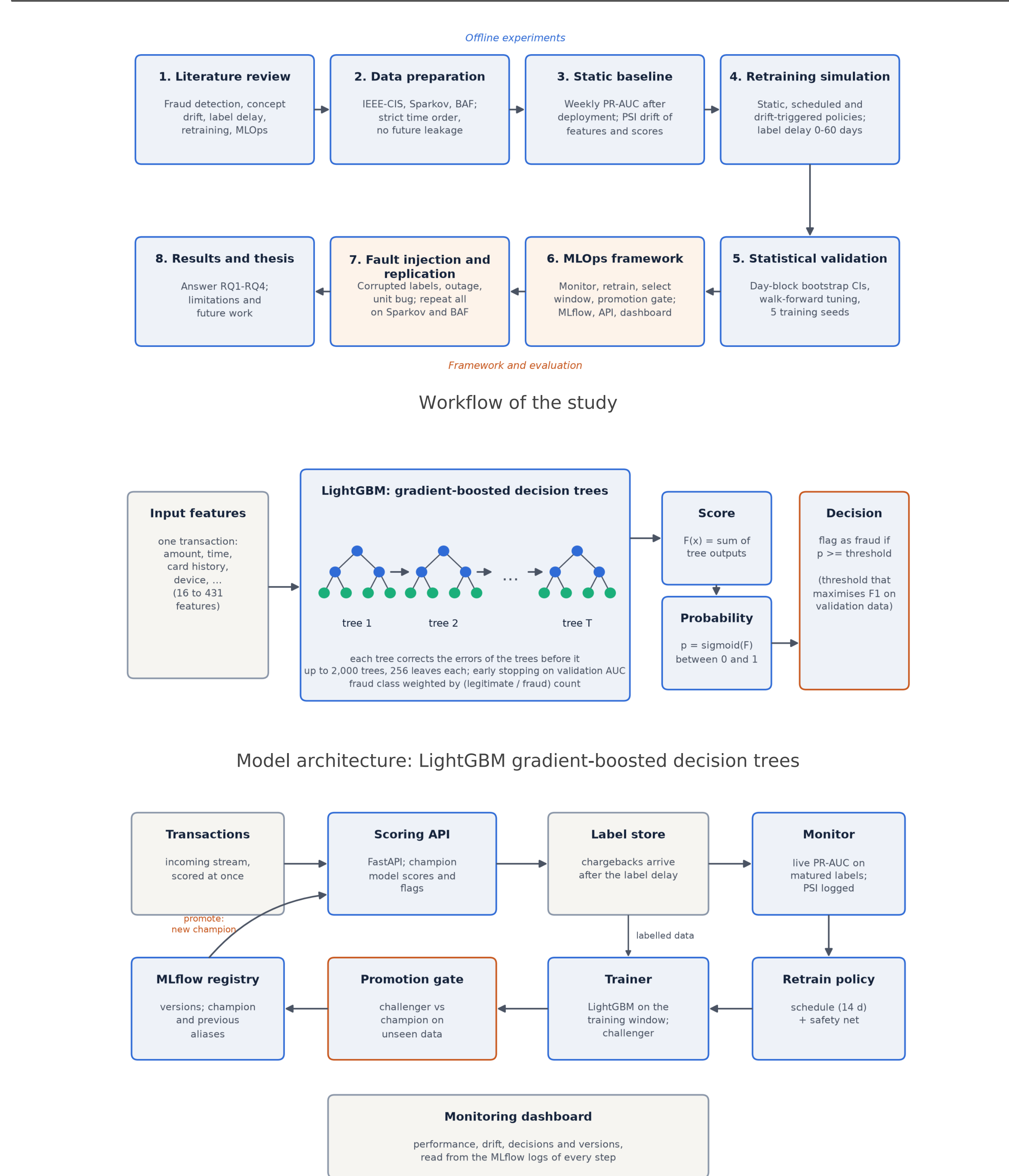
- **IEEE-CIS:** transaction and identity tables joined on TransactionID; text columns as categories; missing values kept.
- **Sparkov:** personal fields removed; cut at 21 Dec 2020 (fraud stops); age, hour, distance and card-history features from *earlier* transactions only.
- **BAF:** -1 codes set to missing; month not used as a feature.
- **All:** sorted by time, never shuffled; no oversampling (fraud class weighted).



output: one table per dataset with TransactionID, TransactionDT (seconds), isFraud and the model features

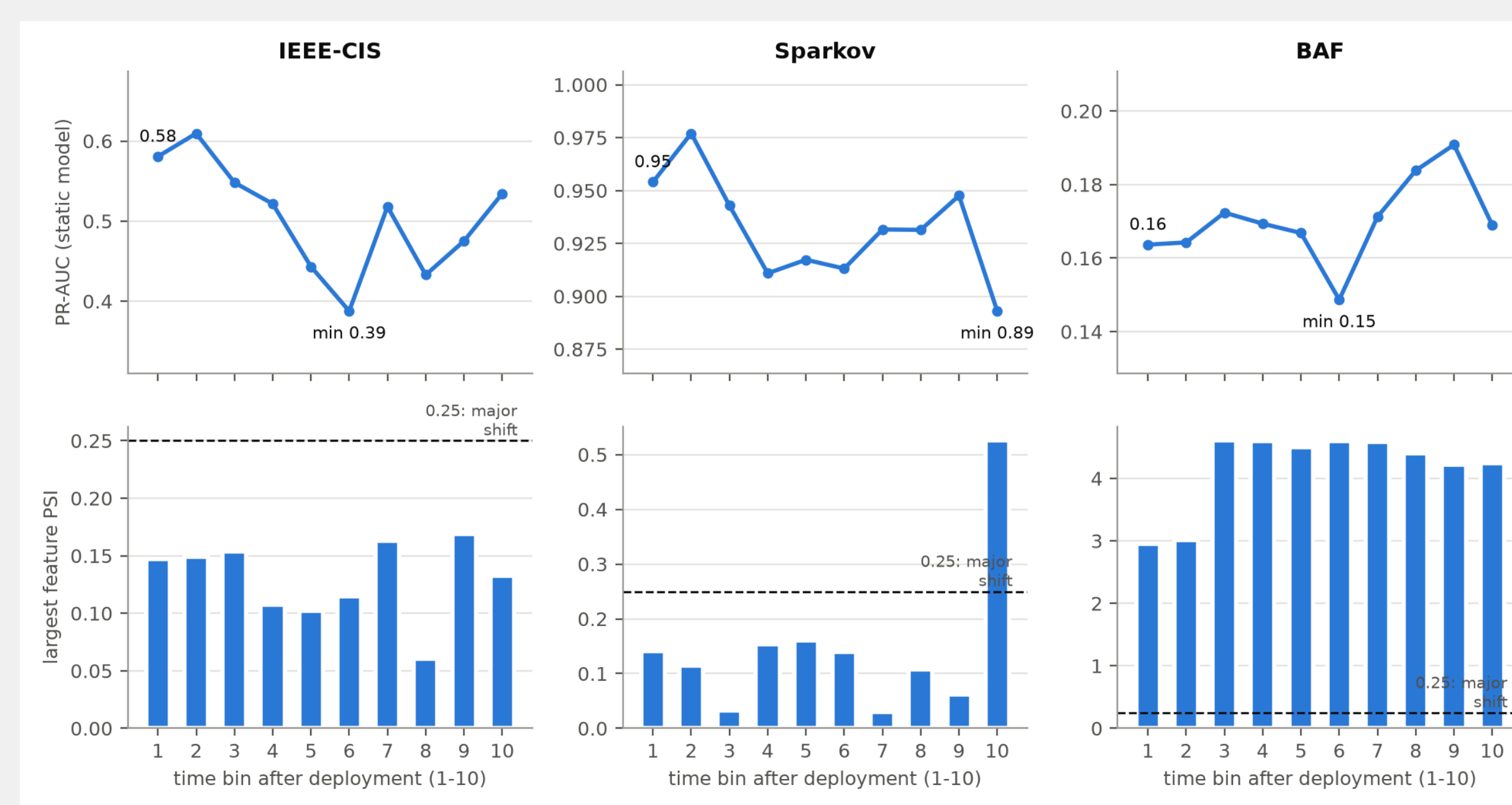
Preprocessing pipeline from raw files to the time-ordered table used by every experiment

Methodology



Proposed framework: each week it monitors, decides, retrains, gates and serves

Result Analysis



Drift measures do not track performance: IEEE-CIS decays while drift stays low; Sparkov and BAF drift without decaying



Conclusion & Future Work

- **Drift indicators can miss real degradation under delayed labels**, while scheduled retraining was more reliable. Here, *drift-aware* means tracking drift through delayed-label performance, not label-free alarms.
- With delayed labels, the schedule matched or beat the drift trigger even at an **equal retraining budget** (IEEE-CIS, BAF): not only from retraining more often.
- **Window selection** and a **promotion gate** make retraining adaptive and safe; results replicated on **two more timestamped datasets** (Sparkov, BAF).
- **Next (supervisor feedback):** equal-budget comparison on Sparkov, and an **API load test** of the scoring service (throughput, p95/p99 latency).
- **Later:** faster investigator feedback, more model families, explanation monitoring and a live deployment.

References

- [1] A. Dal Pozzolo et al., "Credit card fraud detection: A realistic modeling and a novel learning strategy," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, 2018.
- [2] R. S. Menezes and R. H. Filho, "Semantic and structural drift in financial knowledge graphs: A robustness analysis of GNN-based fraud detectors," in *Proc. IEEE ICKG*, 2025.
- [3] M. N. P. Shakil et al., "Feature drift-guided adaptive ML retraining: An MLOps approach for big data analytics," in *Proc. IEEE BigData*, 2025.